

Exercise 4:

1) a) 1. We know from Gauss Law:

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$ and the E -field lines point straight up, giving no flux contribution on the sides,

$$2. \oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

$$2) 2|\vec{E}|A = \frac{\sigma A}{\epsilon_0} \Rightarrow \underline{\underline{E_{\perp} = 2\epsilon_0 \hat{n} \Rightarrow E_{\perp} - E_{\perp} = \frac{\sigma}{\epsilon_0}}}$$

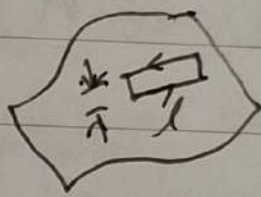
E -field pointing upwards through A

Say positive is up
so both are added

3. The normal component of E is discontinuous by an amount $\frac{\sigma}{\epsilon_0}$ at any boundary.

$$b) 1. \oint \vec{E} \cdot d\vec{l} = \int \vec{E} \cdot d\vec{l} = (E_{above}'' - E_{below}'')l = 0$$

$$\Rightarrow E_{above} = E_{below}$$



Height contribute nothing as $l \rightarrow 0$.

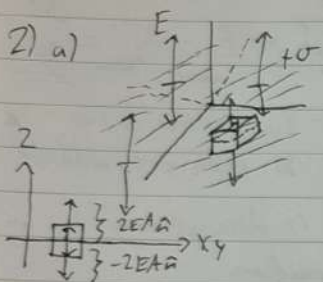
2. When you are moving tangentially to the E -field, it has no effect on you, therefore its components must cancel each other

$$\left. \begin{array}{l} E_{above} = E_{above}'' + E_{above}^+ \\ E_{below} = E_{below}'' + E_{below}^+ \end{array} \right\} E_{above} - E_{below} = E_{above}^+ - E_{below}^+ = \frac{\sigma}{\epsilon_0} \hat{n}$$

d) 1. $V_{above} - V_{below} - \int_L E \cdot dl = 0$ if $L \rightarrow 0$

$\Rightarrow V_{above} = V_{below}$

2. The gradient will inherit E 's discontinuity
so that $\nabla V_{above} - \nabla V_{below} = -\frac{\sigma}{\epsilon_0} \hat{n}$



$\oint E \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0} \quad | \quad Q_{enc} = \sigma A$

$2AE = \frac{\sigma A}{\epsilon_0} \Rightarrow E_{above} = \frac{\sigma}{2\epsilon_0} \hat{n}$

$\oint E \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0} \quad | \quad Q_{enc} = \sigma A$

$-2AE = \frac{\sigma A}{\epsilon_0} \Rightarrow E_{below} = -\frac{\sigma}{2\epsilon_0} \hat{n}$

So that:

$E_{above} - E_{below} = \frac{\sigma}{2\epsilon_0} \hat{n} (1 - (-1)) = \frac{\sigma}{\epsilon_0} \hat{n}$

c) $E_+ - E_- = 0$
 $E_+ + E_- = 2E$
 $E_- - E_+ = 0$
i) $\oint E \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow 2A(E_+ - E_-) = 0$
ii) $-1 - 1 \Rightarrow 2A(E_+ + E_-) = 4AE$

$\Rightarrow E = \frac{\sigma}{4\epsilon_0} \hat{n}$

iii) $-1 - 1 \Rightarrow 2A(E_- - E_+) = 0$

3) a) $W_1 = \frac{1}{4\pi\epsilon_0} q^2 \left(\frac{1}{\sqrt{2}a} - \frac{1}{a} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(\frac{1}{\sqrt{2}} - 1 \right)$

b) $W_4 = \frac{1}{4\pi\epsilon_0} q^2 \left(\frac{1}{\sqrt{2}a} - \frac{2}{a} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(\frac{1}{\sqrt{2}} - 2 \right)$

c) $W_2 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} (-1) \quad W_1 = 0$

$W_{tot} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} (-1 + \frac{1}{\sqrt{2}} - 1 + \frac{1}{\sqrt{2}} - 2)$

$W_{tot} = \frac{1}{2\pi\epsilon_0} \frac{q^2}{a} \left(\frac{1}{\sqrt{2}} - 2 \right)$

4) a) $\nabla \cdot E = -\Delta V$

$\Rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2 V}{\partial \varphi^2}$

$\Rightarrow \frac{20}{r^4 \sin\theta \cos\theta} + \frac{20(\cos^2\theta + \sin^2\theta)\cos\theta}{r^4 \sin\theta} - \frac{10 \cos\theta}{r^4 \sin\theta}$

$\Rightarrow \frac{20}{2^4} - \frac{10}{2^4} - \frac{10}{2^4} = 0$

b) $W = q \cdot (V(b) - V(a)) = 0.010 (0.7 - (-5)) = 0.053$
 $= 53 \text{ mJ}$

5) a) The formula $W = \frac{\epsilon_0}{2} \int E^2 dz$ is an approximation to the formula

$W = \frac{\epsilon_0}{2} \left(\int E^2 dz + \oint V E \cdot da \right)$ where we integrate over a gaussian sphere. By making the sphere larger and towards all space, the flux integral over the surface goes towards 0, and we can shorten the formula

b) Electrostatic energy (W) does not obey superposition principle because it is quadratic in the fields. You will get "cross terms" in compound systems:

$$W_{\text{tot}} = \frac{\epsilon_0}{2} \int E^2 dz = \frac{\epsilon_0}{2} \int (E_1 + E_2)^2 dz$$

$$\Rightarrow \frac{\epsilon_0}{2} \int E_1^2 + E_2^2 + 2E_1 \cdot E_2 dz$$

$$\Rightarrow W_1 + W_2 + 2 \frac{\epsilon_0}{2} \int E_1 \cdot E_2 dz$$

where if $E_1 = E_2 \Rightarrow W_{\text{tot}} = 4W_1$

This is different to Coulomb's law that does follow superposition, but is used more for point charges.

c)  $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$ $E^2 = \frac{1}{(4\pi\epsilon_0)^2} \frac{q^2}{r^4}$ $E=0$ inside

$$W = \frac{\epsilon_0}{2} \frac{q^2}{(4\pi\epsilon_0)^2} \int \frac{1}{r^4} dz = \frac{q^2}{32\pi^2\epsilon_0} \int \frac{1}{r^2} d\phi d\theta dr$$

$$W = \frac{q^2}{8\pi\epsilon_0} \int \frac{1}{r^2} dr = \frac{q^2}{8\pi\epsilon_0 R}$$

$$V = - \int E \cdot dl = - \frac{q}{4\pi\epsilon_0} \int \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0 R}$$

$$W = \frac{1}{2} q \sigma V_b = \frac{q}{8\pi\epsilon_0 R} \int \sigma da = \frac{q^2}{8\pi\epsilon_0 R}$$

It's the same! $\therefore$